SWATI RANI

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Motivated and detail-oriented Computer Science Engineering student seeking an entry-level role to apply strong Java, SQL, and backend development skills. Eager to contribute to enterprise-grade projects through problem-solving, collaboration, and continuous learning.

EDUCATION

NIST Institute of Science and Technology — Berhampur, Odisha
BTech, Computer Science Engineering,8.31 — 2021-2025
PRANAVANANDA VIDYA MANDIR — Agartala, Tripura
Higher Secondary (XII), CBSE,84.5% — 2019 – 2020
PRANAVANANDA VIDYA MANDIR — Agartala, Tripura
Matriculation (X), CBSE, 85.8% — 2017 – 2018

EXPERIENCE

OIL AND NATURAL GAS CORPORATION (ONGC) - **Summer Trainee** — May 2024 – June 2024

Dispensary Management System

Java, JDBC, MySQL, NetBeans (Java Swing)

- Developed a desktop application to manage over **1,000+ patient records** improving data accessibility by **60%**.
- Designed and integrated 8+ GUI screens using Java Swing for user-friendly interactions.
- Optimized SQL queries to improve data handling performance and consistency.

ACADEMIC PROJECTS

Web Quiz App

HTML, CSS, JavaScript

- Designed and developed a responsive **web-based quiz application** using HTML, CSS, and basic JavaScript.
- Implemented interactive **multiple-choice questions**, dynamic question loading, and **real-time score tracking**.
- Added **answer validation** with user feedback and a final results summary on completion.

Email Spam Detection

Python, Scikit-learn, Pandas, Naive Bayes, Logistic Regression

- Developed a machine learning model to classify emails as spam or not spam with ~96% accuracy using Naive Bayes
- Trained and tested the model on a dataset of over 5,000 labeled emails.
- Compared performance of Naive Bayes and Logistic Regression, improving precision by 10% through feature tuning

Cognitive Situational Analysis and Predictive System

Python, CNN, SVM, Random Forest, EEG, Scikit-learn

- Built an ML-based system to classify human emotions (positive, negative, neutral) from EEG signals with **91% accuracy using a CNN model.**
- Processed and analyzed over **10,000 EEG signal samples** across multiple channels and frequency bands.
- Compared the performance **of 4 models (CNN, SVM, Random Forest, Gradient Boosting)**; CNN outperformed with highest precision and F1-score..

CERTIFICATES

- Problem Solving and Competitive Coding — June,2024
- Database Management System (NPTEL) — March,2024
- Python for Data Science (NPTEL) — July 2023-August 2023
- Advanced C programming and Data Structure — March,2023

POSITION OF RESPONSIBILITY

COORDINATOR

NIST COUNSELING SERVICE, NIST — 2022 – 2025

Guided and counseled first year students to help them navigate the challenges in their college life and organized events to create a supportive environment.

SKILLS

- **TECHNICAL SKILLS:** Java (Core Java, OOPs, Collections, Multithreading, Exception Handling), C, Python, SQL
- **FRAMEWORK AND TOOLS:** JDBC, Java Swing, Maven, NetBeans, MySQL, Git, Spring(basic-currently learning), Hibernate(basic),REST API(basic),Intellij,Eclipse
- **Web Technologies:** HTML, CSS, JavaScript
- **DOMAINS:** Data Structures and Algorithms, RDBMS, Operating Systems, Machine Learning